INTERVIEW PREPARATION GUIDE

Kenera International Trading PLC

Junior Network Engineer — Interview Readiness Guide

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WHAT'S INSIDE

Behavioral Q&A • OSI Model • TCP vs UDP • Life of a Packet
DHCP • DNS • NAT • VLAN • STP • OSPF • ARP
Structured Troubleshooting • Firewalls • Cisco Command Reference

Prepared using publicly known networking fundamentals and the candidate's own project experience.

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1 Tell Me About Yourself

MODEL ANSWER

Thank you for giving me this opportunity. My name is Eyasu Kerie, and I recently graduated with a Bachelor's degree in Electrical and Computer Engineering from Addis Ababa University, with a specialization in Computer Engineering.

During my studies, I completed an internship in Network and Computer Engineering at INSA, where I gained practical exposure to network devices, computer systems, troubleshooting, and IT infrastructure. I also worked as a Data Encoder on Ethiopia's National ID project, which built my attention to detail, data accuracy, and teamwork.

Beyond networking, I enjoy software development. I've built several projects, including an ERP system using React, FastAPI, PostgreSQL, and JWT authentication — work that helped me understand how networking supports modern enterprise applications.

I'm eager to begin my professional career as a Junior Network Engineer, where I can keep learning technologies such as Cisco networking, routing, switching, and network security while contributing to the company's success.

2 Why Do You Want to Work at Kenera?

This question is almost guaranteed — answer with specific, genuine reasons.

MODEL ANSWER

Kenera has built a strong reputation as one of Ethiopia's leading ICT companies. I understand the company provides enterprise networking, cybersecurity, cloud, and infrastructure services across many organizations.

I'm particularly interested because Kenera works closely with global technology partners such as Cisco and Fortinet. Working here would let me learn industry best practices while working with enterprise-grade networking equipment.

I believe this environment will help me grow technically and professionally, and I'm excited to contribute my networking knowledge, problem-solving ability, and willingness to learn.

Tip: Before the interview, confirm Kenera's actual service lines and any named technology partners on their website — swap in the real specifics so this answer reads as researched, not generic.

3 The OSI Model

Layer	Function	Example
7 — Application	User-facing services	Browser, email client
6 — Presentation	Encryption & compression	SSL / TLS
5 — Session	Creates & manages sessions	NetBIOS
4 — Transport	End-to-end delivery (TCP/UDP)	Segments
3 — Network	Logical addressing & routing	Router
2 — Data Link	Physical (MAC) addressing	Switch
1 — Physical	Cabling & raw signal transmission	Hub / cable

MEMORY AID

> “Please Do Not Throw Sausage Pizza Away” — Physical, Data Link, Network, Transport, Session, Presentation, Application (bottom to top).

4 TCP vs UDP

MODEL ANSWER

> TCP is connection-oriented. Before any data moves, it establishes a connection through the three-way handshake — SYN, SYN-ACK, ACK — and guarantees reliable delivery through acknowledgments, sequencing, retransmission, and flow control.
>
> UDP is connectionless. It sends data without establishing a connection first and does not guarantee delivery, ordering, or error recovery. Because it carries less overhead, it's faster and better suited to real-time traffic.

Uses TCP	Uses UDP
HTTP / HTTPS	VoIP
FTP	Video streaming
SSH	Online gaming
Email (SMTP)	DNS queries

5 Life of a Packet — Full Walkthrough

Scenario: a user types www.google.com into a browser. This is one of the hardest questions asked — answer it as a clean, ordered sequence.

1. Cache check — The browser first checks whether Google's IP address is already cached locally.

2. DNS resolution — If not cached, the computer sends a DNS request to resolve the domain name into an IP address.

3. ARP lookup — If the destination is on another network, the computer uses ARP to learn the MAC address of the default gateway.

4. Encapsulation — The request is encapsulated layer by layer: Application builds the HTTP/HTTPS request, Transport adds the TCP header, Network adds IP addresses, Data Link adds MAC addresses, and Physical converts it into electrical or wireless signals.

5. Switching — The switch forwards the frame based on the destination MAC address.

6. Routing — The router forwards the packet based on the destination IP address.

7. TCP handshake — Before data is exchanged, TCP performs the three-way handshake: SYN, SYN-ACK, ACK.

8. Request sent — The HTTPS request travels to Google's server.

9. Server response — Google processes the request and returns the webpage.

10. Rendering — The browser receives the data, verifies integrity via TCP, decrypts the HTTPS traffic, and renders the page.

6 DHCP (DORA Process)

MODEL ANSWER

> When a client joins a network, it broadcasts a DHCP Discover message. The DHCP server replies with a DHCP Offer containing an available IP address. The client sends a DHCP Request asking to use that address, and the server confirms with a DHCP Acknowledgment, which also carries settings such as the subnet mask, default gateway, and DNS server.
>
> **Memory aid: DORA** — Discover, Offer, Request, Acknowledge.

7 DNS

DNS translates human-readable domain names such as google.com into IP addresses. Without it, users would need to remember numerical addresses for every site. DNS typically uses UDP port 53 for standard queries and TCP port 53 for larger responses or zone transfers.

8 NAT & PAT

NAT allows multiple devices on private IP addresses to share one or more public IP addresses. It conserves IPv4 address space and hides internal addressing from the public internet. PAT (Port Address Translation) extends this by letting many devices share a single public IP, distinguishing each connection by port number.

9 VLAN Segmentation

A VLAN is a logical segmentation of a physical network that groups devices together regardless of their physical location. Benefits include stronger security, reduced broadcast traffic, easier management, and better performance. Access ports carry traffic for a single VLAN, while trunk ports carry traffic for multiple VLANs using IEEE 802.1Q tagging.

10 Spanning Tree Protocol (STP)

STP prevents Layer 2 switching loops by placing redundant links into a blocking state. It elects a Root Bridge and calculates the shortest loop-free path for every switch. If the active path fails, STP can activate a backup path to maintain connectivity.

11 OSPF Routing

OSPF is a dynamic link-state routing protocol. Routers exchange Link-State Advertisements to build a complete map of the network and use Dijkstra's algorithm to calculate the shortest path. OSPF uses cost as its routing metric, sends Hello packets to discover neighbors, and requires all routers to connect through Area 0, the backbone area.

12 ARP

ARP (Address Resolution Protocol) maps an IPv4 address to a MAC address within a local network. When a host knows a destination's IP but not its MAC address, it broadcasts an ARP request; the device holding that IP replies with its MAC address, enabling Data Link layer communication.

13 Troubleshooting: No Internet Connection

Scenario: a user reports they cannot connect to the internet. Walk through this layer by layer to show structured, methodical thinking.

Layer 1 — Physical — Check power, cables, Wi-Fi connection, and link lights.

Layer 2 — Data Link — Verify switch port status, VLAN assignment, and MAC address learning.

Layer 3 — Network — Check IP configuration with ipconfig/ifconfig; confirm valid IP, subnet mask, gateway, and DNS. Ping 127.0.0.1, then the local IP, then the default gateway, then an external address such as 8.8.8.8.

DNS check — If IP connectivity works but websites fail to load, run nslookup google.com to verify name resolution.

Based on where the results break down, isolate whether the issue sits with the client, the switch, the router, the ISP, or DNS.

14 Firewalls — Stateless vs Stateful

A firewall monitors and filters network traffic based on security rules to protect systems from unauthorized access. A stateless firewall evaluates each packet independently against predefined rules. A stateful firewall tracks the state of active connections and makes decisions based on the context of the communication, giving stronger security with less rule complexity.

15 Behavioral: A Difficult Technical Problem

Draws on the ERP project — anchor behavioral answers in real, verifiable experience.

MODEL ANSWER

> During my ERP project, I ran into a deployment issue where the frontend worked but the backend couldn't connect properly to the PostgreSQL database, due to environment variable and configuration problems. I worked through it systematically — reviewing logs, verifying environment variables, correcting the database configuration and deployment settings, then testing the API endpoints with Postman until everything worked. It reinforced how valuable structured troubleshooting, careful log analysis, and layer-by-layer verification are before assuming where a problem actually lives.

16 Strengths

- Fast learner
- Strong analytical thinking
- Solid troubleshooting skills
- Team player
- Adaptable to new technologies
- Eager to learn Cisco and Fortinet enterprise networking

17 Areas for Growth

> One area I keep improving is balancing perfection with efficiency. Early in my projects I sometimes spent extra time refining small details before moving on. I've since learned to prioritize tasks, meet deadlines, and improve features iteratively while still protecting quality.

18 Why Should We Hire You?

MODEL ANSWER — CLOSING

> I believe I'm a strong candidate because I bring a solid academic foundation in computer engineering, practical experience from my internship, hands-on experience building software systems, and a genuine interest in networking. I'm eager to learn, adapt quickly, and contribute to the team — and if given the opportunity, I'll work hard to grow my skills and provide reliable support for Kenera's network infrastructure and customers.

R1 Cisco Command Quick Reference

Show / Diagnostic Commands

```
show ip interface brief
show running-config
show vlan brief
show interfaces
show mac address-table
show ip route
show version
show startup-config
copy running-config startup-config
```

Basic Configuration Example

```
enable
configure terminal

hostname SW1

interface FastEthernet0/1
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

vlan 10
name SALES

interface FastEthernet0/2
switchport mode access
switchport access vlan 10
```

R2 Final Interview Strategy

For a Junior Network Engineer role, structured thinking matters more than memorizing every command. For any technical question, walk through it in this order:

- **1.** What the technology is
- **2.** How it works
- **3.** Why it's used
- **4.** A practical example from experience

Interviewers generally value a logical troubleshooting process and clear communication as much as memorized facts. With an academic background in computer engineering, INSA internship experience, and hands-on software projects, the goal is to present this knowledge with confidence and method — not to recite it.

Before the interview: confirm Kenera's current services, clients, and any named technology partners so the "why this company" answer reflects real, up-to-date specifics rather than assumptions.